CAPSTONE PROJECT

Music Playlist & Listening-Stats Manager

PRESENTED BY

CHANDANA SREE Roll no. 2620030040

WAFA MEHAJABEEN Roll no. 2620030428

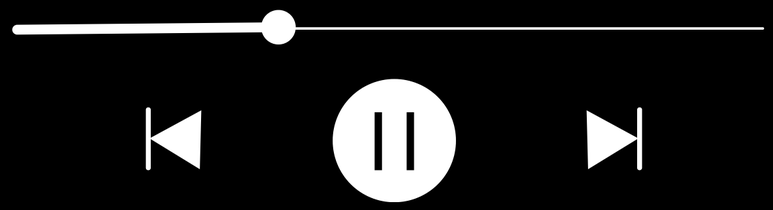
Section-2

Search

Abstract

The Music Playlist and Listening Stats Manager is a digital system designed to make personal music management simple and organized. It brings playlist creation, song management, playback tracking, and listening analysis together in one platform. The system stores music information and tracks listening activity to give users a clear view of their music collection and listening habits.

Instead of treating each song separately, it connects songs, playlists, and listening data to identify useful patterns and preferences.
The project combines data management, logical processing, and user interaction to create a practical music-management solution. It shows how everyday listening data can be organized and converted into meaningful information, making music management more systematic, useful, and engaging.

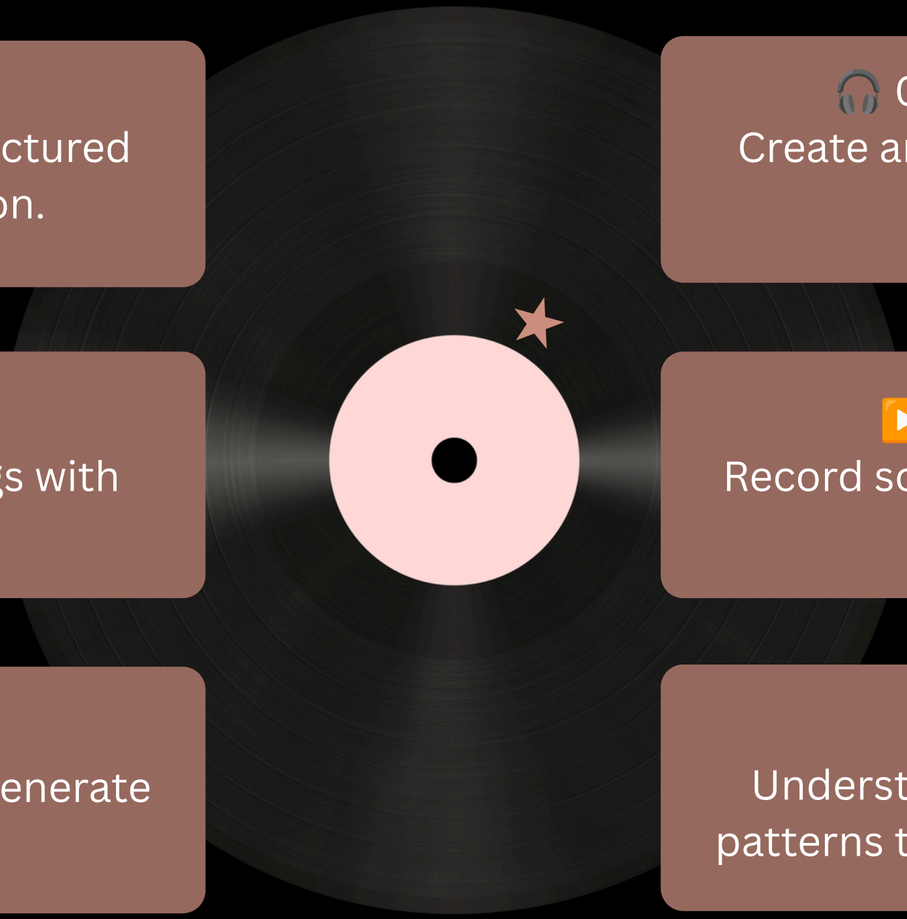


Introduction



The Music Playlist and Listening Stats Manager is a Java-based system for managing songs, playlists and listening activity. It provides a structured way to store music information, manage playlists and analyse listening behaviour. The project was initially developed using basic Java concepts and was later extended with object-oriented programming, collections, file handling and functional programming concepts.

PROJECT OBJECTIVES



🎵 01 — ORGANIZE
Organize songs into a structured digital music collection.

🎧 02 — PERSONALIZE
Create and manage personalized playlists.

🔎 03 — DISCOVER
Search and retrieve songs with ease.

▶️ 04 — TRACK
Record song plays and listening activity.

📊 05 — ANALYZE
Process listening data to generate useful statistics.

💡 06 — INSIGHT
Understand individual listening patterns through meaningful data.

The Genres Of Music

ROCK



POP



R&B



HIP-HOP



CLASSIC



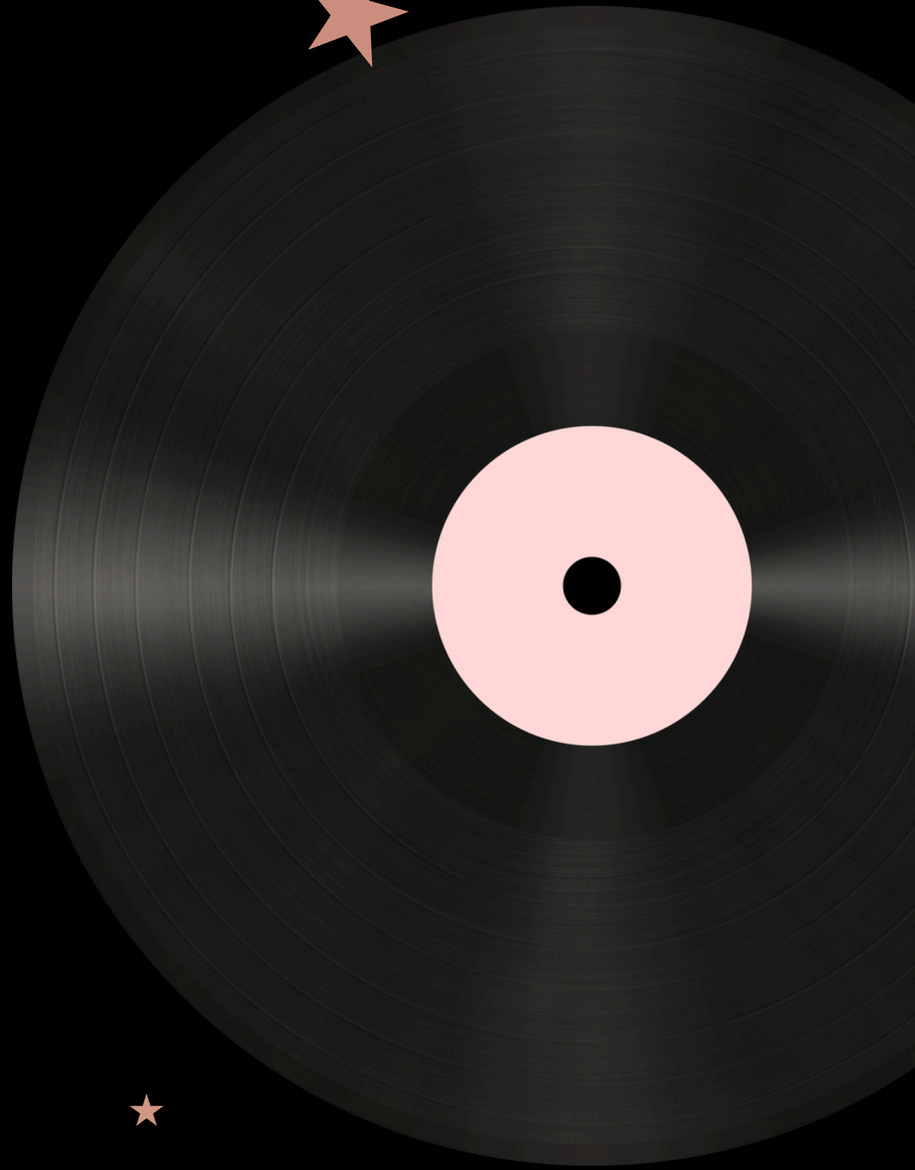
Advantages & Limitations

ADVANTAGES

- Simple and easy to operate.
- Handles music data in an organized manner.
- Supports flexible data management.
- Preserves information using file storage.
- Makes analysis of listening data easier.
- Demonstrates practical use of Java concepts.

LIMITATIONS

- Currently designed as a local Java application.
- File-based storage is less scalable than a database.
- No graphical user interface.
- No online music streaming.
- Recommendations and analysis are limited to the implemented logic.



Project Development

Development approach and concepts applied

The Music Playlist & Listening Stats Manager was developed as a Java-based console application for organizing songs, creating playlists, and analyzing listening activity. The project gradually integrates basic programming concepts with advanced Java features.

- CO1: Basic Java programming concepts, variables, data types and operators
- CO2: Control statements, loops, switch and problem-solving
- CO3: Arrays, methods, Scanner and data handling
- CO4: Inheritance, polymorphism and custom exceptions
- CO5: Records, file handling and HashMap
- CO6: Streams, filtering and listening-data analysis

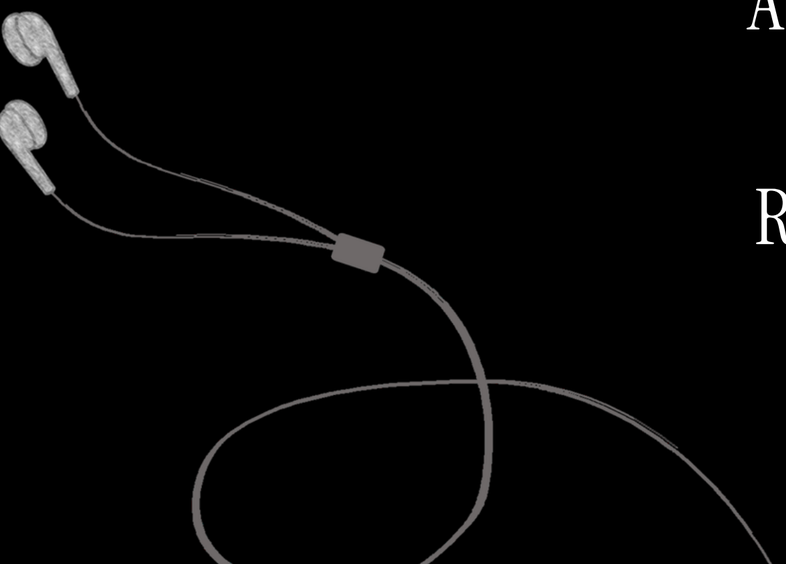
Topics Covered

Core Java

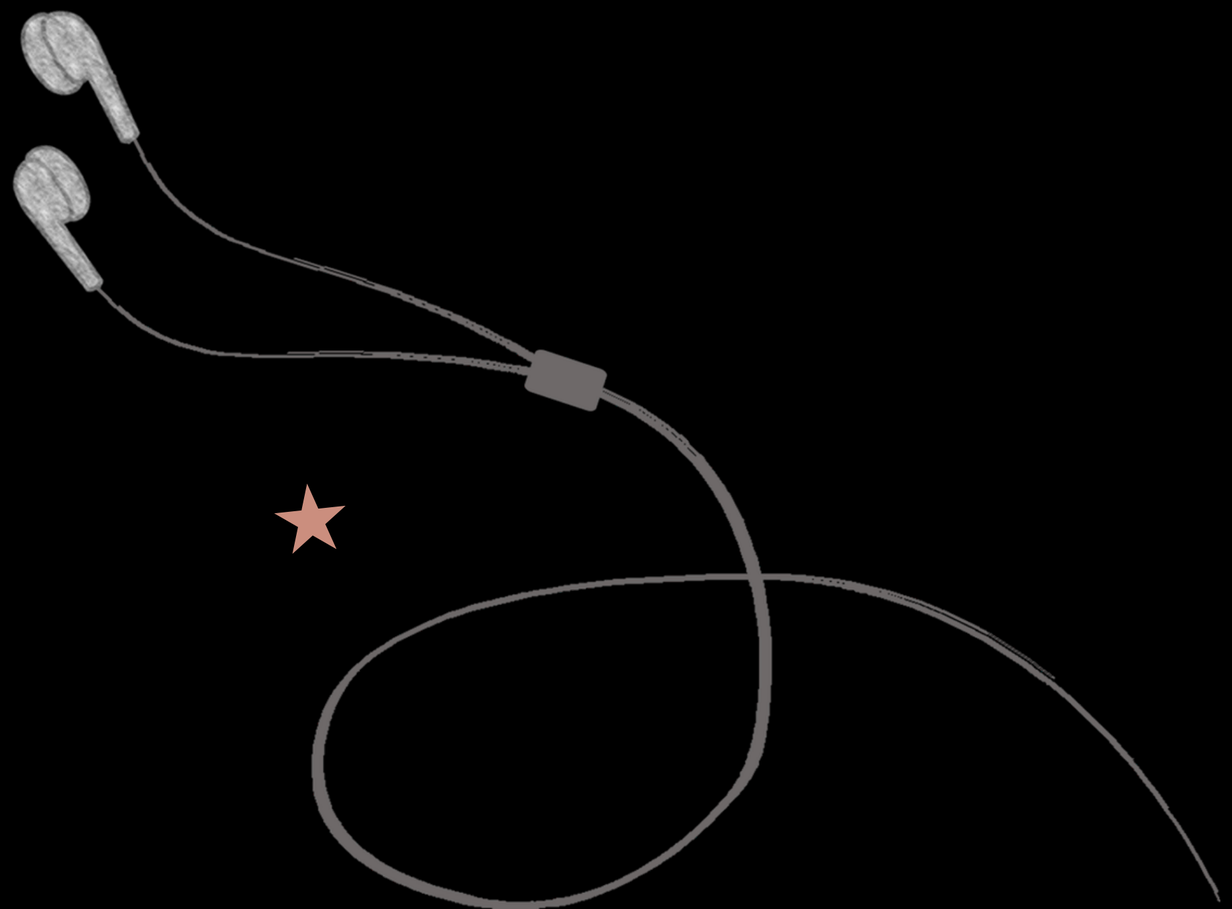
Variables & Data Type, Operators, If-Else & Switch, Loops, Arrays, Methods, Scanner Input,

Advanced Java Features

Abstract Classes, Inheritance, Polymorphism, Custom Exceptions, Records, File Handling, HashMap, Streams & Filtering



System Workflow :



START

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DISPLAY MUSIC PLAYLIST MANAGER MENU

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MANAGE SONGS

- VIEW SONGS
- SEARCH SONG
- ADD SONG

↓

MANAGE PLAYLIST

- ADD SONGS TO PLAYLIST
- VIEW PLAYLIST

↓

PLAY SONG

- UPDATE PLAY COUNT
- STORE LISTENING DATA

↓

LISTENING STATISTICS

- TOTAL PLAYS
- MOST PLAYED SONG
- ADVANCED STREAM-BASED ANALYSIS

↓

SAVE NEW SONG INFORMATION

↓

RETURN TO MENU

↓

EXIT

Future Scope

Add permanent storage for the complete music library

Support larger song collections and multiple playlists

Add playlist editing and song removal features

Provide richer listening analytics and visual reports

Add sorting and ranking of songs based on listening activity

Develop a graphical interface for easier interaction

Conclusion

The Music Playlist & Listening Stats Manager provides a simple way to organize songs, create playlists and track listening activity. The project demonstrates the gradual use of Java programming concepts, from basic arrays and methods to exceptions, object-oriented features, file handling, HashMap and Streams. It provides a practical foundation that can be further expanded into a more complete music management system.



Thank You

